

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE
SUBJECT : CYBER SECURITY

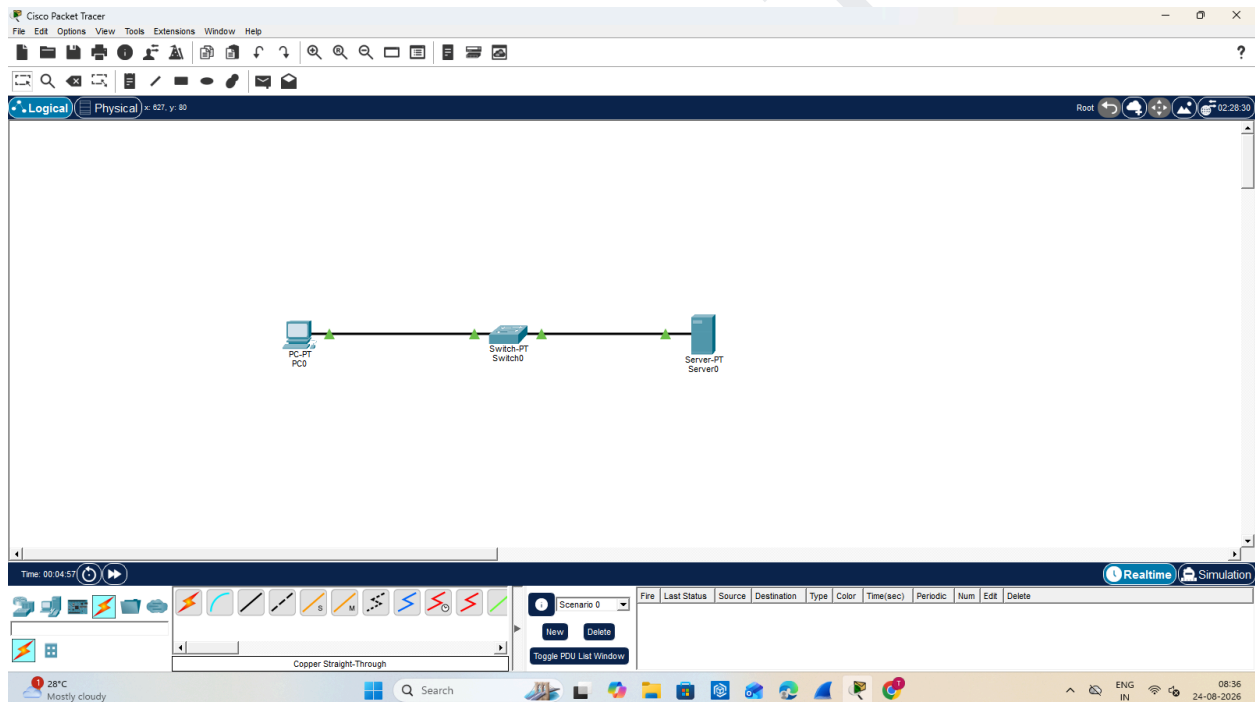
AIM : Web Security with SSL/TLS

Configure and implement secure web communication using SSL/TLS protocols, including certificate management and secure session establishment.

Objective

To configure and implement secure web communication using HTTPS (SSL/TLS concept) between a client PC and a web server in Cisco Packet Tracer.

Network Topology



Name : Shubham Sanjay Karape
Roll No : T085

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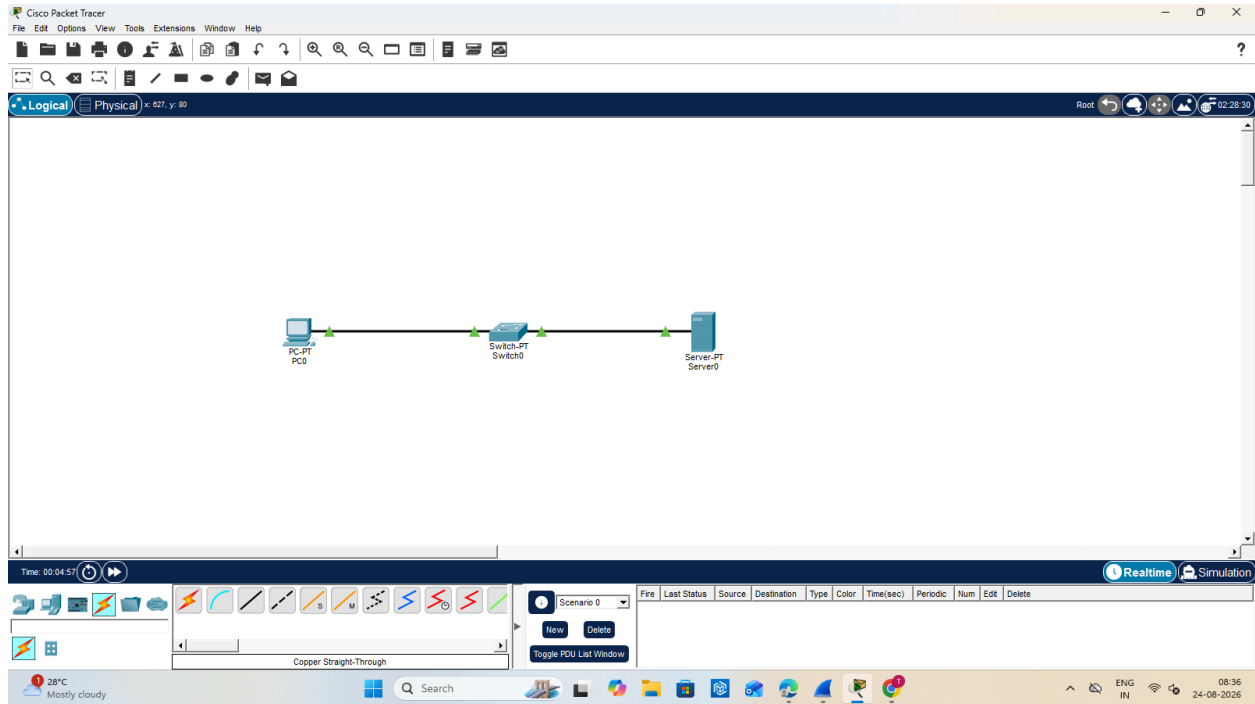
Step 1: Open Cisco Packet Tracer

Step 2: Add a PC

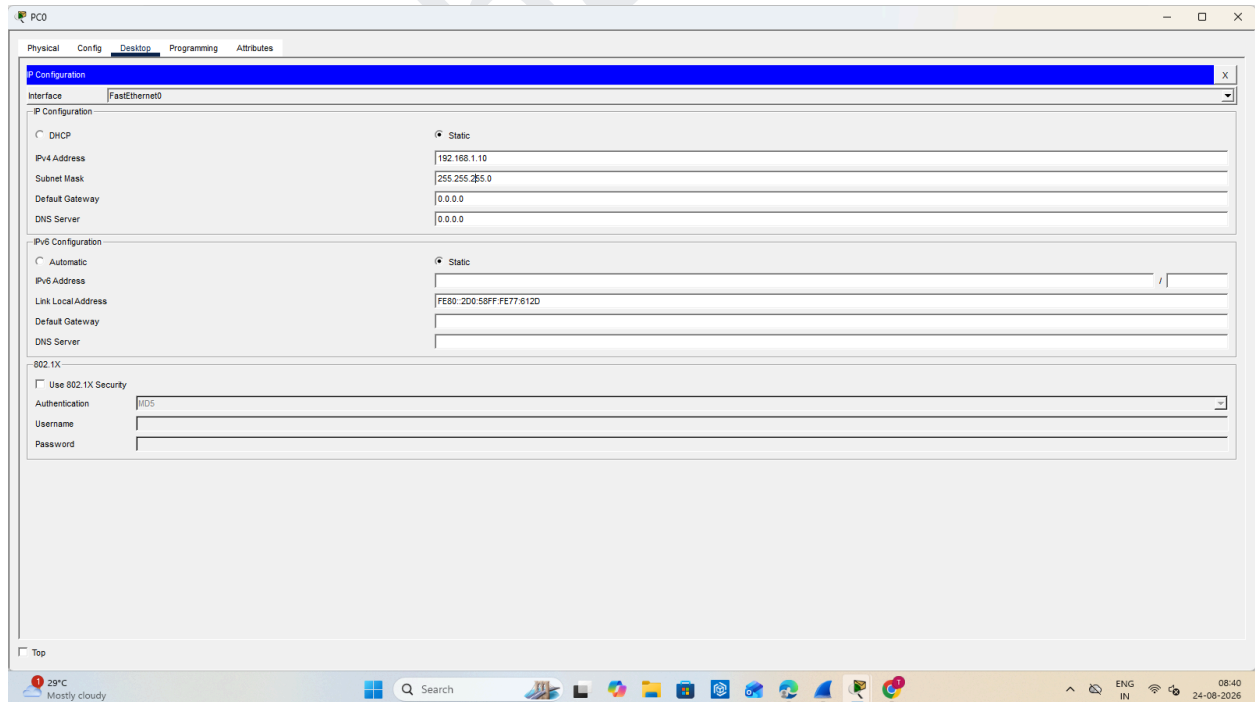
Step 3: Add a Server

Step 4: Add a Switch

Step 5: Connect the Devices



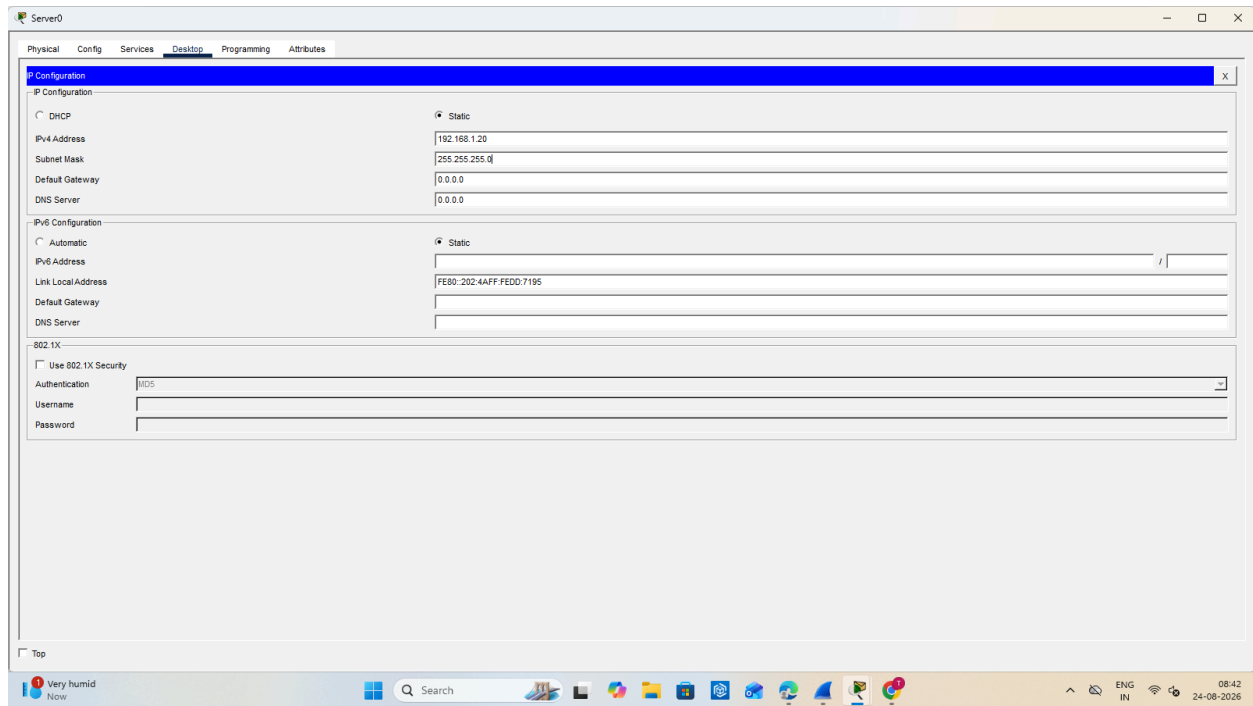
Step 6: Configure IP Address on PC



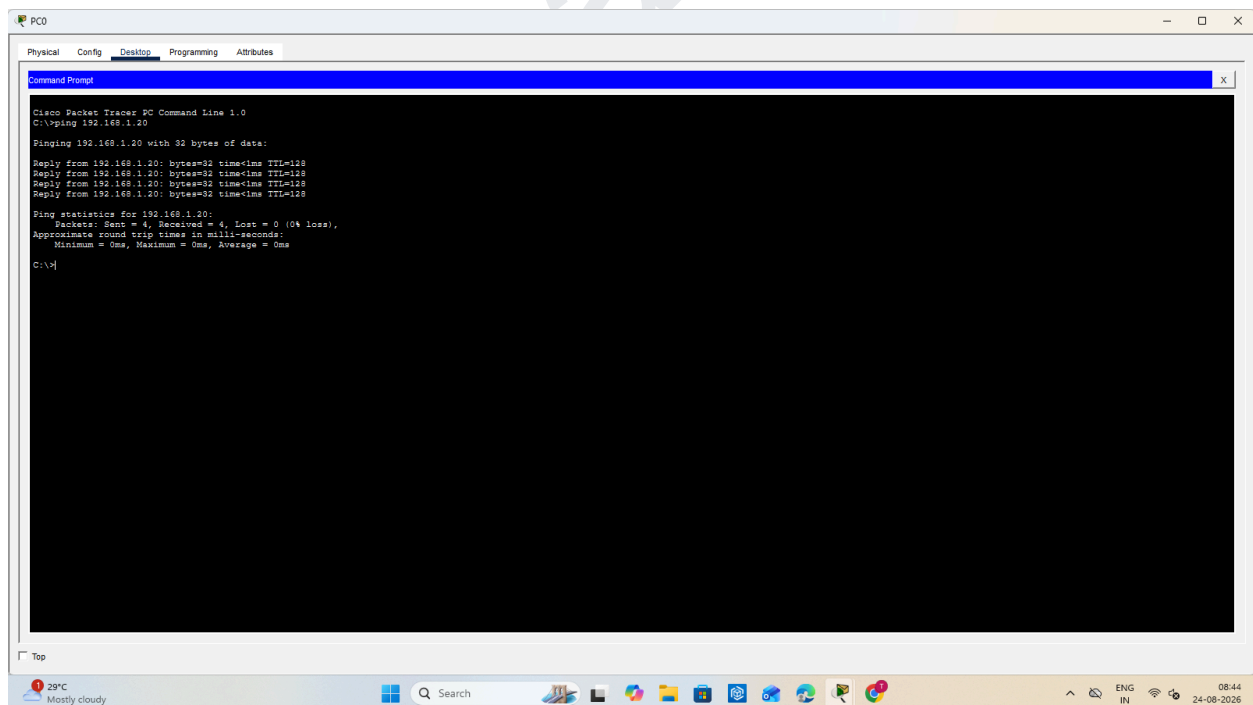
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Step 7: Configure IP Address on Server



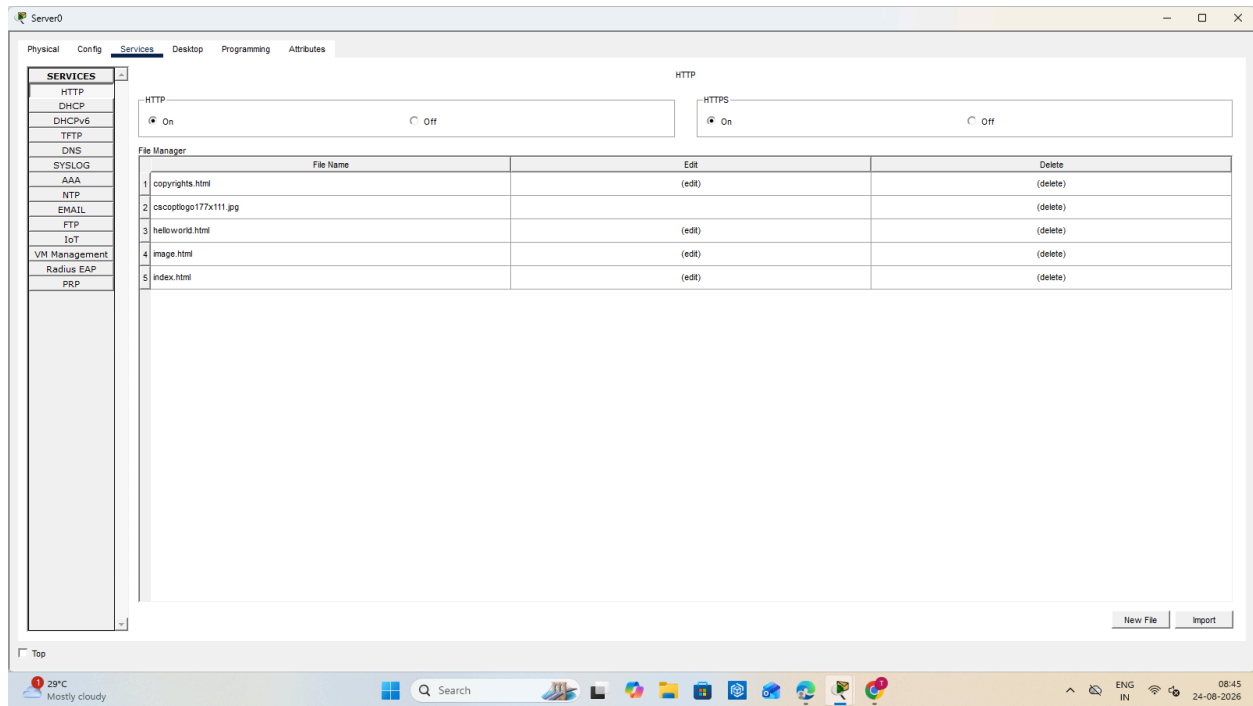
Step 8: Test Network Connectivity



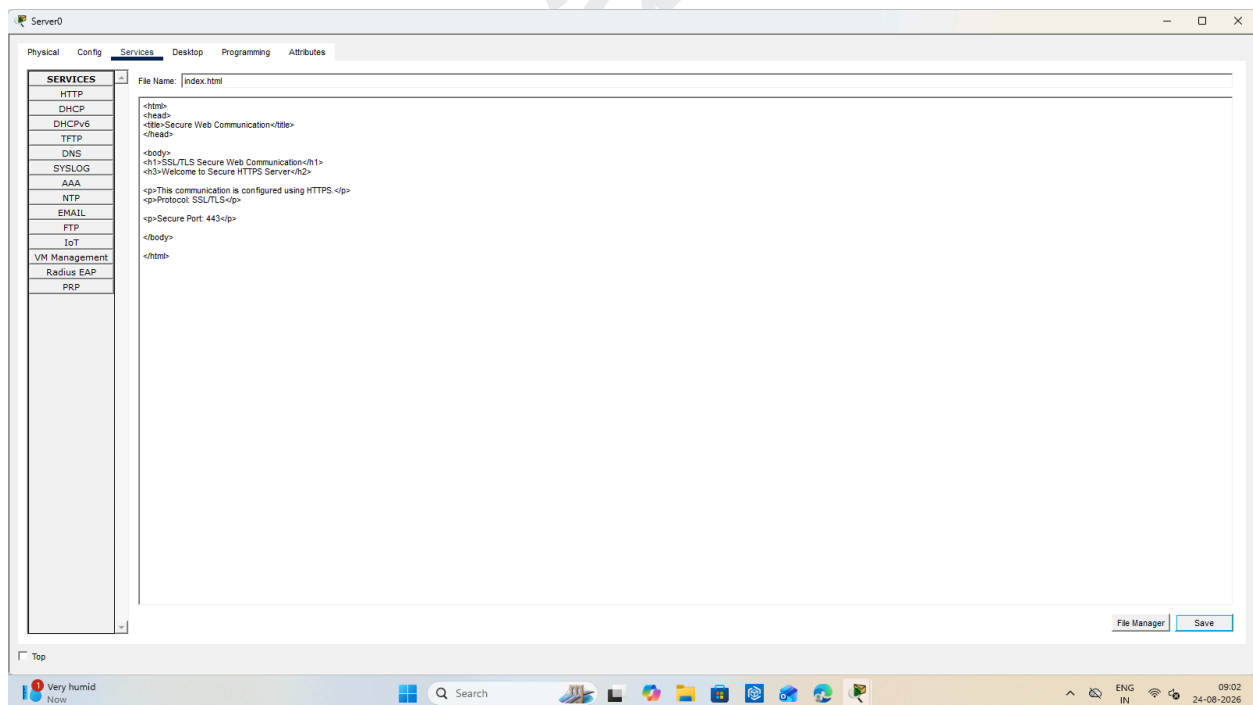
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Step 9: Configure HTTP and HTTPS Service



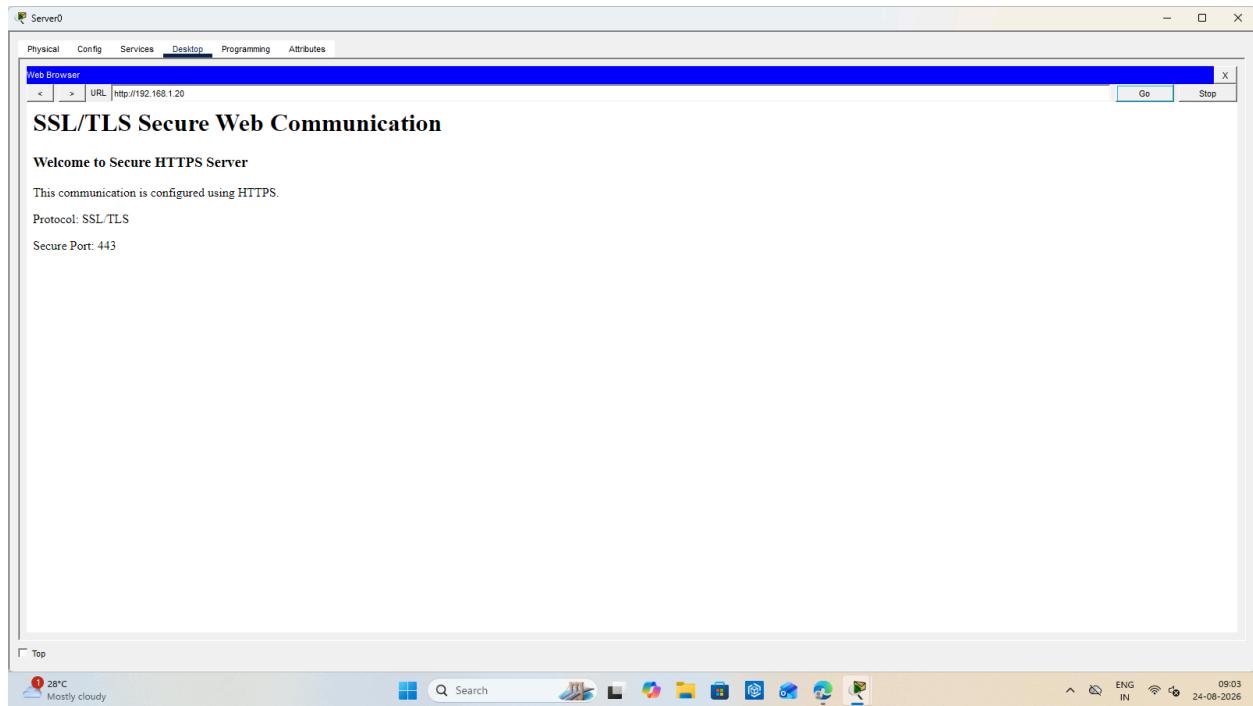
Step 10: Configure a Web Page



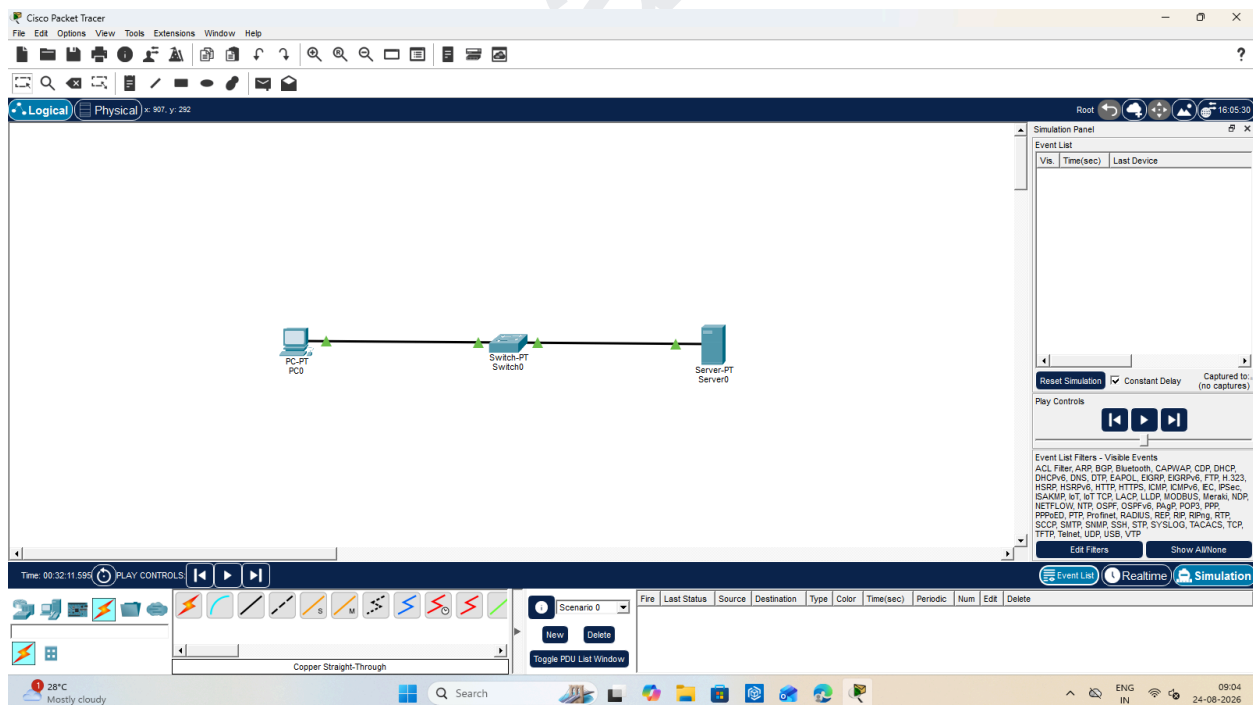
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Step 11: Test Normal HTTP Communication



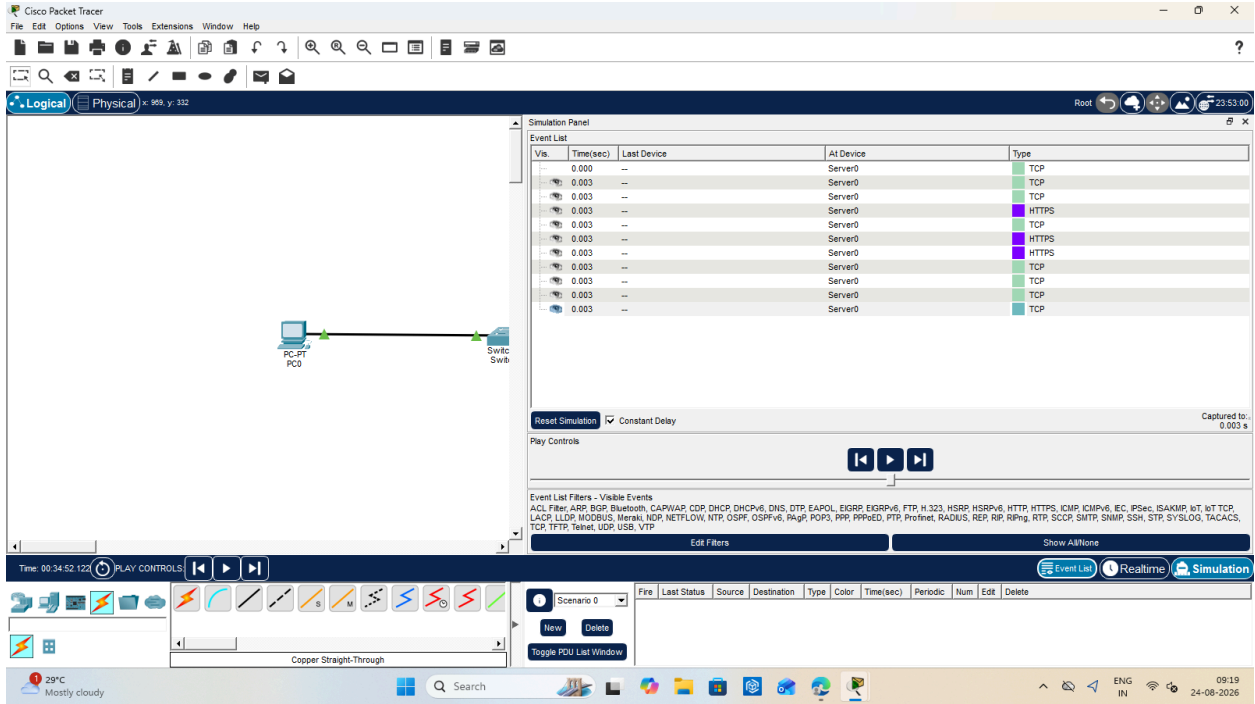
Step 13: Use Simulation Mode



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Step 14: Observe HTTP Communication



The screenshot shows the Cisco Packet Tracer interface. The main workspace displays a network topology with a PC (PC-PT PC0) connected to a switch (Switch SW0). The Event List panel on the right shows a series of events, including HTTP and TCP communication between Server0 and the switch. The events are listed in a table with columns for Time(sec), Last Device, AI Device, and Type.

Time(sec)	Last Device	AI Device	Type
0.000	--	Server0	TCP
0.003	--	Server0	TCP
0.003	--	Server0	TCP
0.003	--	Server0	HTTPS
0.003	--	Server0	TCP
0.003	--	Server0	HTTPS
0.003	--	Server0	HTTPS
0.003	--	Server0	TCP
0.003	--	Server0	TCP
0.003	--	Server0	TCP
0.003	--	Server0	TCP

The interface also includes a Simulation Panel with a Reset Simulation button, Play Controls, and a list of Event List Filters. The bottom status bar shows the time as 00:34:52.122 and the temperature as 29°C.